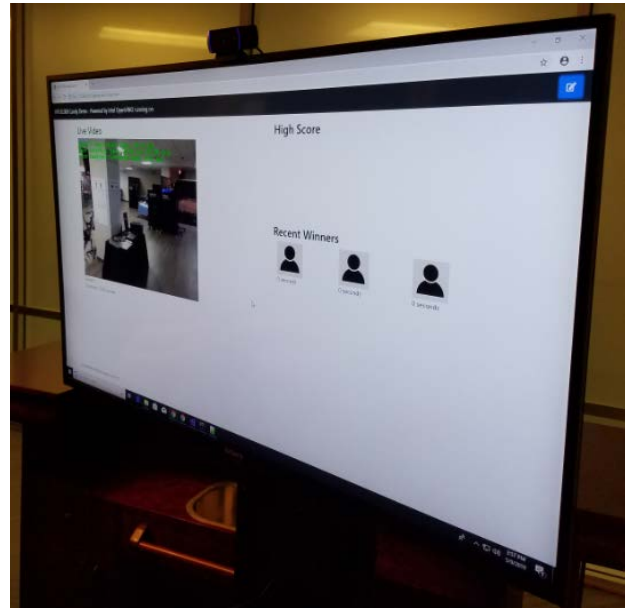


EL300 Happy Candy Demo (Version 8)

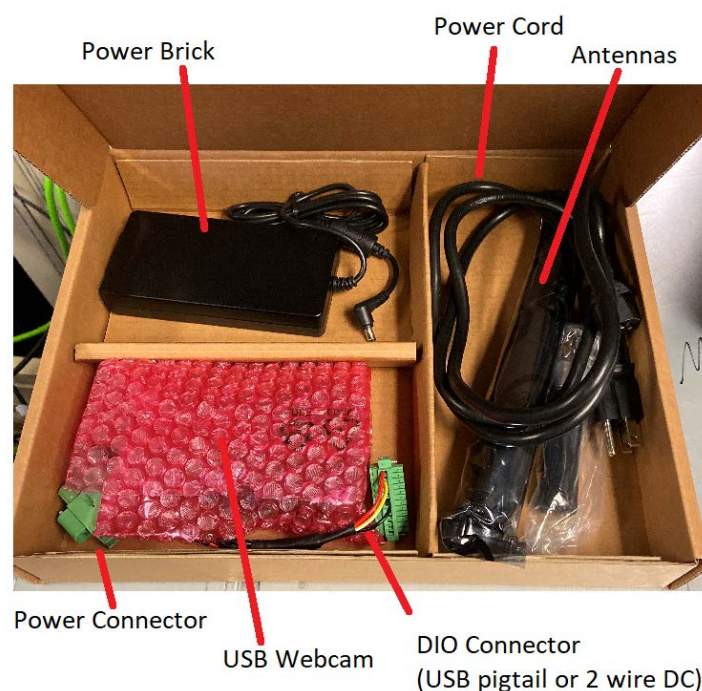


Description of the Demo

This demo uses a web camera attached to the HPE Edgeline EL300 with digital IO and the HPE Edgeline VPU module. It does video inferencing with the VPU (Vision Processing Unit) module with Intel's OpenVINO framework. The models do facial detection and sentiment detection. That is then further processed to see if the user is "happy" for a particular duration. If they are then the application will use the DIO to dispense candy.

Contents

There should be a box with the Candy Dispenser and a box with the EL300. Inside the box with the EL300 there is a smaller box that includes multiple components.



Setting Up the Demo

- 1) Power up the EL300 to the OS and connect the monitor and USB keyboard/mouse (or wireless keyboard/mouse). Network is not needed unless you want to demonstrate the iSM as well or remotely connect to the unit.
- 2) Plug in the webcam and situate it so that it can get a good view of a face. A tripod is also included in the box.
- 3) Plug in DIO connector to EL300.
- 4) Plug connector from DIO to candy machine. There are two versions of hardware connections.
 - a. DC Jack (Rev 1)
 - i. You only need to plug the DIO connector in as well as the USB camera.



- b. USB connection (Rev 2)
 - i. You must plug in the DIO connector and the male USB to the top right USB port. Then plug in the USB webcam to another USB port.
 - ii. Plug the candy dispenser into the female USB pigtail that connects to the DIO connector. If you plug the dispenser into a normal USB port, and the candy dispenser power switch is on, it will continuously run. This is useful if you are trying to fully empty out the dispenser.
 - iii. You must also move the candy dispenser switch from the top position (Off) to any other position. A green LED will turn on under where the candy dispenses to indicate power is good. If no LED is seen, verify you have Rev 2 hardware that plugs in via USB, that the USB is plugged in, and the switch on the back of the candy dispenser is in any position but Off.



- 5) Login to Windows OS with username "iot" or "HPE" and password "password".

Running the Demo

The demo is simple to run.

For version 7 and later:

- On the desktop, there is a “demo_vX” folder where X is the version of the demo. Inside there is a “launch_candy_demo.bat”. Double click that and a Chrome browser show pop up in full screen mode. Minimize any other windows. Press F11 to get out of Chrome full screen mode if needed.

For versions 1-6:

- On the desktop, there is a “demo” folder. In that folder you can double click on either the “run_demo_cpu.bat” which will run the demo on the CPU or the “run_demo_vpu.bat” which will run the demo on the VPU. After you run the demo you must double click the index.html to open it up. The demo will not actually run until you do this. All options can be modified with the blue button at the top right. Below is an example of what the UI looks like. The lines on the screen indicate where the face detection will happen. This is to help eliminate multiple facial detections which confuse the demo.

Options

Found in the top right corner behind the blue button, there are a few items you can change. Change the settings you want, hit the apply button and then close the window.

- 1) Candy duration. This is how long the candy motor will operate.
- 2) Identification Line Opacity. This is how bright the lines show up on the screen.
- 3) Identification Line Color. The color can be modified due to different backgrounds and locations.
- 4) Identification Window Size. This is how narrow or wide the face identification area will be.
- 5) Statistics Color. These are the outputs from the candy_demo that describe how long different aspects of inference required. You can also turn these off.

Possible Issues

- 1) DIO gets stuck on.
 - a. Make sure to unplug the M&M dispenser quickly or kill and restart the demo program.
- 2) Candy dispenser not working.
 - a. Make sure the DIO is connected and Candy Dispenser is connected via DC jack or USB cable.
 - b. If Rev 2 (USB plug), make sure the USB pigtail male portion is plugged into the EL300 and the switch on the candy dispenser is in any position except Off. The green LED in the front of the candy dispenser should be on.
 - c. If Rev 1 (DC Jack), make sure the batteries are installed and are working.
 - d. Make sure there is only one instance of the demo running at one time. It may help to reboot the machine.
 - e. Make sure people are happy and the happy countdown reaches 0.

Detailed Theory of Operation

This demo is based on the “interactive_face_detection_demo” from Intel’s OpenVINO framework. The base demo will only do face detection on the CPU. With extra arguments you are able to add features such as emotion detection (which is used in this demo) and others such as head pose/orientation and gender detection. You can also change the targets that each of those options target with the -d option. -d just changes the face detection target while -d_em changes the target for the emotion detection. That is why you are able to modify these in the batch file. These also require models to run. The interactive face detection demo that we run though is customized and

includes hardcoded paths to the necessary models. The changes to the demo code should be on the HPE github. After compiling the file, rename the output to candy_demo.exe and move to the bin folder of the demo.

The DIO board is used to control the candy dispenser. The DLLs for the Microchip USB to I2C adapter (MCP2221) are required for this demo. They are in the proper location but keep in mind if you are compiling the solution you do need to link them properly if that is not already done in the solution. When you run the program the DLL must be present. The Microchip DLL is used instead of OTLink since OTLink is still more difficult to use as an add-on when this was first created. The primitives for sending I2C commands are in the source code. All GPIOs are configured as outputs and all bits are written to turn the GPIOs on (sinking current) by writing 0x00 and then off by writing 0xFF.

The candy dispenser is modified internally. There are 3 connectors inside the candy dispenser. One handles the LED and power from batteries, one handles the IR sensor, one goes to the motor. The internals are reworked so that battery power goes straight to the motor positive side via a soldered wire. Then the positive side of the DC jack that connects to the DIO is used on the motor negative side. The negative side of the DC jack is connected to the battery GND wire in the candy dispenser. This allows the DIO board to sink the current by connecting the other side of the motor to GND. The DIO board can support 500mA per output and in one measurement with a multi-meter the current was only about 20mA which was surprisingly low.

The demo uses a webpage to show the video output. The images from OpenCV in the candy_demo app are being encoded to base64 and transmitted over websockets (via websocketd) to the webpage. Javascript takes care of the duration of happy status.

Demo Software Revision History

Version 1 – Initial version for lab testing

Version 2 – Added more flags for testing/configuration. Added middle zone where a face must be to eliminate two people standing side by side which confuses the demo, thus person must be in the middle of the frame. Made full screen the default option and added the HPE logo. Happy countdown is always present.

Version 3 – Interim between 2 and 4

Version 4 – Web page is main interaction with program. Offers longest winner and previous winner history.

Version 5 – Interim between 4 and 6

Version 6 – Interim between 5 and 7

Version 7 - Support framework for supporting multiple endpoints. Only CPU supported now. CPU is default, no option to select a different inference engine. Added emotion smoothing. Changed deployment to be "launch_candy_demo.bat" method which automatically launches everything.

Version 8 - CPU is default but option to select engine exists. Changed to OpenVINO 2019.2.275 which includes emotion smoothing already. Changed countdown timer to 0.1 resolution to look less choppy. Added closest face detection to eliminate issues with multiple faces in the FOV.

Demo Hardware Revision History

Revision 1 – This uses a DC jack to connect just a control to the motor using a common ground. This requires batteries and did not use the built in switch for the candy dispenser.

Revision 2 – This version uses a USB pigtail on the DIO. The male USB goes to the EL300 to power the candy dispenser. This means there are no batteries. The female USB on the pigtail is what the candy dispenser plugs into. There is also an LED on the front of the candy dispenser that will turn green when there is power to the candy

dispenser. The candy dispenser switch is utilized now. Without it on, the LED will not be green and the dispenser will not work.